15 Prove algebraically that the difference between the squares of any two consecutive odd numbers is always a multiple of 8

Let n be an integer. Consecutive odd numbers:

$2n - 1$ and $2n + 1$.

1 mark
General odd-number expressions.

	$2n$	1
$2n$	$4n^2$	$2n$
1	$2n$	1

	$2n$	-1
$2n$	$4n^2$	$-2n$
-1	$-2n$	1

$(2n + 1)^2 - (2n - 1)^2$

$= (4n^2 + 4n + 1) - (4n^2 - 4n + 1)$

1 mark
Correctly expand both squares and subtract.

$= 8n$

Since n is an integer, $8n$ is a multiple of 8.

1 mark
The algebra proves it for every integer n.

(Total for Question 15 is 3 marks)